Acute Lymphoblastic Leukemia Classification Algorithm

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Abstract

Acute Lymphoblastic Leukemia (ALL) is the most common form of pediatric cancer. It requires fast and precise diagnostics to ensure an effective treatment. Manual microscopic examination of bone marrow samples or blood smears remains the clinical gold standard for ALL diagnosis to date. Nevertheless, this method is limited by human fatigue and it may lead to misdiagnosis. This project explores the application of Deep Learning (DL) in the medical field, specifically using the ResNet50 architecture and Transfer Learning to create a model for the classification of leukemic lymphoblasts. A two-phase training strategy was implemented by first freezing the backbone of the ResNet50 architecture and then applying fine-tuning to the whole architecture. The final fine-tuned model achieved an overall accuracy of 83.28% and a high precision of 85.04% detecting leukemic cells. While specificity reached a value of 96%, the 55.82% value for the ALL class sensitivity highlights the room for improvement the model has and the complexity in distinguishing complex morphological features. This project is primarily designed as an educational roadmap to investigate and test the skills I have acquired about DL and Convolutional Neural Networks over the last months. As a Biomedical Engineering student at the *Universitat de Barcelona (UB)*, the motivation for this study lies in the integration of Deep Learning in the healthcare sector.

Keywords: Acute Lymphoblastic Leukemia, ResNet50, Transfer Learning, Biomedical Engineering, Deep Learning.

1. Introduction

Acute Lymphoblastic Leukemia (ALL) is a rare fast-growing and aggressive cancer that starts in the bone marrow, causing it to produce large numbers of immature, non-functional white blood cells called lymphoblasts [1]. The risk of developing ALL reaches its maximum in children younger than 5 years of age, from that age to the age of 20 this risk decreases a lot, the risk of developing ALL starts to slowly increase again after the age of 50. Overall, a little less of half of the cases of ALL are in adults [2].

For many decades, manual microscopic examination of blood and bone marrow samples has been the principal way to diagnose ALL. This method had lots of limitations most of them related to human error factors. First of all, lymphoblasts (early-stage immature cells) look extremely similar to lymphocytes (healthy mature cells) from a morphological perspective, the difference between these two cells can be appreciated in the chromatin's texture/pattern and the nucleus-cytoplasm ratio. Therefore, to the human eye this slight changes between the two types of cells can very often be passed by leading to erroneous diagnosis. Moreover, long shifts, stress and many hours looking at cells in the microscope can produce ocular fatigue leading to even more misdiagnosis. Last but not least, doing manual diagnosis requires much more time than diagnosis made by medical devices/softwares [3][4].

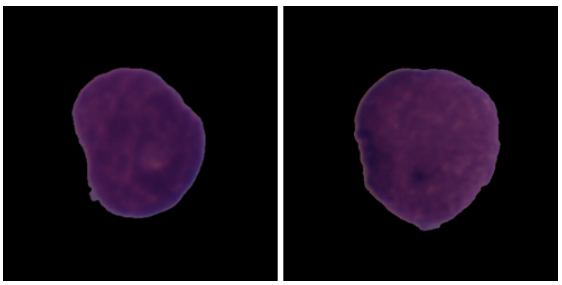

Figure 1: Comparison to highlight the similarity of a cancerous and a healthy cell. The cancerous ALL cell (lymphoblast) is the one on the left and the healthy cell (lymphocyte) is the one on the right.

Recently, because of all the limitations that manual diagnosis has, doctors have started to implement Deep Learning (DL) algorithms such as Convolutional Neural Networks (CNN) to help with the diagnosis of ALL. It is important to highlight that artificial intelligence is not meant to replace doctors in these diagnosis tasks but to be a magnificent tool for them to use. Among the advantages of using an algorithm for this task we can appreciate its consistency, algorithms do not get tired meaning they are able to analyze as many images of cells as needed and their performance will be consistent. Machine Learning/ Deep Learning models are also able to detect much more details from an image than the human eye, they can find pixel-level patterns and textures which would be impossible for the naked eye to see. Finally, an algorithm can do the job much faster than a person ever could manually [3][4][5]. As I mentioned before, AI should be used as a tool and not as a replacement of a doctor, ideally an algorithm used for ALL diagnosis should flag the cells that have a big probability of being cancerous for the doctor to check them later.

2. Related Work

Before getting into specific technical aspects of my model I wanted to acknowledge the work done by many others who have published their work in this health-tech field. This block is dedicated to explaining the models and architectures that have inspired me to build my model and also to mention the most relevant diagnostic algorithms that are currently being used. As stated in the "*Abstract*" this paper and the model I have build have education purposes, improving my programing skills and my intuition and knowledge about CNNs. Therefore, I do not expect to add any value to the field of diagnostics through computer algorithms but instead to learn along the journey.

2.1. Residual Networks (ResNet)

Residual Netwoks architecture was introduced by Kaiming He et al. in 2015. What made these networks revolutionary is the fact that they allowed neural networks to be much deeper than before without losing performance in the training set as happened before. The innovation of ResNet comes from the *Skip Connections* (or *Shortcut Connections*) which allowed information from the network to skip some layers passing the original information forward [6].

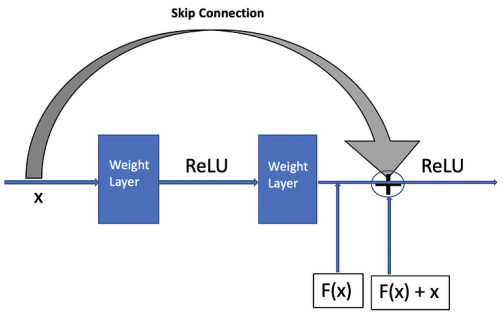

Figure 2: Diagram of Skip Connections in ResNets [7].

This made it much easier for the model to focus on tiny differences instead of re-learning basic shapes in every layer, it allowed for deeper feature extraction, in our case helping to differentiate between a healthy and a leukemic cell as the differences are in the small details.
I will go into more detail about ResNets in "*3. Model Architecture and Training Procedure*" as it is the architecture I used in my model.

2.2. U-Net and Medical Image Segmentation

The U-Net was developed by Ronneberger et al. in 2015. While the ResNet architecture is very useful for classification tasks such as detecting whether a cell is healthy or not, the U-Net has become the most popular architecture for image segmentation in the medical industry. This architecture is specifically designed to work with small medical datasets. The U-Net architecture is not as good as classifying as the ResNet but it tells us with pixel-level precision which pixels have been recognized as the "tumor" in the cell, this information can be extremely useful for biomedica researchers or doctors as the architecture "explains" by drawing a heat-map over the image where the abnormalities that triggered the cancer detector are [8].

3. Model Architecture and Training Procedure

While Residual Networks advantages provide a strong foundation for image classification, applying this architecture to an Acute Lymphoblastic Leukemia (ALL) Classificator required many modifications for the algorithm to handle specific morphological details and differences in images of hematological cells.

In this section, the development of the classification algorithm has been detailed from end-to-end. This includes the preprocessing of the C-NMC_Leukemia dataset to ensure computational efficiency, modifications made to the ResNet50 architecture to satisfy the necessities of the classification and the explanation of a two-phase training strategy used to focus on improving sensitivity.
The following sections address the main challenges of extracting distinguishing features from highly similar cell populations.

3.1 Dataset and Preprocessing

The dataset used for training this model is the *C-NMC_Leukemia* dataset. It consists of segmented images of cells from real microscopic images. These images are representative of images in the real-world as they have some illumination and noise errors. Nevertheless, most of these errors have been fixed by *"Larxel"*, the dataset creator. The dataset has been uploaded from *Kaggle* and it contains a total of 15.135 images from 118 patients with two labeled classes: *all* for leukemic cells and *hem* for healthy normal cells [9].

3.1.1 Data Acquisition and Structuring

This dataset was accessed via the *kagglehub* library. Data in this dataset is organized into training, validation and testing folders. Because of an issue trying to read the csv file which contained the labels in the validation folder, the decision to only work with the data in the training_data folder was made, this training_data folder had three subsets of folders inside containing labeled images. Therefore, the training set was defined by concatenating two of the three subsets and the validation and test sets were defined by splitting the last of the three folders. The final data split was:

- Training Set – 7.094 images
- Validation Set – 1.777 images
- Test Set – 1.776 images

3.1.2. Image standardization

The ResNet50 architecture utilized in this algorithm required a fixed input size. The original images were 450 x 450 pixels RGB images, these were resized to 224 x 224 pixels as this is the image size of images in the *ImageNet* dataset. This is important as Transfer Learning from a pre-trained

ResNet50 with ImageNet images has been used as a base model for the algorithm. Finally, images have also been normalized using *.preprocess_input* a built in function in *tensorflow* which among others, divides by 255 the pixels values and centers the data for each RGB channel.
This standardization is crucial to accelerate gradient descent convergence and ensure that the model treats all features with equal numerical importance.

3.1.3. Data Augmentation

Data Augmentation is a technique used to artificially increase the size and diversity of the training set. It is very important in the medical field where datasets are usually smaller. The decision to implement this technique has been made because of the reduced size of the dataset. As explaiend in "*3.1.1. Data Acquisition and Structuring"* the data used to train the model was less than the original dataset provided. As cells are orientiation-independent and these can have multiple sizes and shapes the following three modifications were made to artificially increase the training data size: Random flips (mirroring) over the vertical and horizontal axes, Random rotations in the following range [-72°, 72°] and Random zoom, zooming images in and out by 20%.
This technique significantly reduces overfitting and helps to the overall performance of the algorithm.

3.1.4. Data Distribution and Class Imbalance

After developing the first versions of the model and struggling with the sensitivity results a small data distribution analysis was made. Within the training set 4.815 images belong to the ALL class and 2.279 belong to the HEM class.
An approximate 2.1:1 ratio can be appreciated in this distribution, this shows a great class imbalance where the malignant cells outnumber by a lot the healthy cells.
This distribution favoured the majority group (malignant cells) in the first models built, achieving high accuracy values. The problem was that sensitivity was low and most of the mistakes the algorithm made were by not detecting leukemic cells as cancerous which would be fatal in a real medical environment. To solve this issue the 2:1 ratio was taken into account and the implementation of Class Weights was made correct this.

3.2. Model Architecture

The architecture developed for this project is designed to build a specialized medical diagnosis classifier from a general purpose image recognition algorithm. By implementing Transfer Learning, the model utilizes the ResNet50 architecture as the backbone to identify pixel-level cellular features while also reducing the computational cost of training.
The following sections clarify the utilization of the ResNet50 as the base model architecture with Transfer Learning and the modifications made to design an algorithm able to identify malignant lymphoblasts.

3.2.1. Feature Extraction

The backbone of the built architecture is based on ResNet50 model. This decision to use a residual network as the core of the model is based on the utilization of residual blocks, which implement skip connections to allow gradients to flow through the network more effectively and solve the vanishing gradient problem. This allows for the creation of deeper neural networks which are able to learn more complex and detailed features without losing performance as they get deeper. Because of the high detail needed to differentiate a healthy

cell from a leukemic one, this was the chosen architecture for the ALL classifier.

The model was initialized with pre-trained weights on the ImageNet dataset. This allowed the network to be immediately trained to recognize low-level features (edges, patterns, textures…) which are also important in biological contexts, reducing training time and data volume needed for the model to learn to detect this features.

The ResNet50 has a fully-connected layer at the end of the architecture, this layer was prepared for classifying 1000 different categories including animals, vehicles, and objects. The final layers, including the dense layer, were removed (`include_top = False`) to build this model as modifications had to be made to adapt the model to an Acute Lymphoblastic Classifier but transforming the backbone structure of the ResNet50 into a pure feature extractor.

3.2.2. Preprocessing and Data Augmentation Layers

To ensure consistency and robustness, the preprocessing and data augmentation stages were added into the model architecture. A Lambda layer and Data Augmentation layers have been used to do this.

The Lambda layer is implemented with the `.resnet50.preprocess_input` function which normalizes every input image as required by the original ResNet50 architecture. The inclusion of this preprocessing function eliminates any issue related to different data formats being used during the evaluation of the model and its training.
The Data Augmentation layer was defined with the `data_augmentation()` function previously created. As explained in "*3.1.3. Data Augmentation*" this function artificially increases the diversity of training data in each epoch by implementing random rotations, flips and zooms to the original images of the dataset.

3.2.3. Dimensionality Reduction

The ResNet50 backbone generates high-dimensional feature maps which have to be reduced in dimensionality to be suitable for classification. To do this, Global Average Pooling (GAP) was implemented as a layer with the `.GlobalAveragePooling2D()` function.

In traditional CNNs a Flatten layer is typically used. The Flatten layer reduces dimensionality by concatenating every spatial activation within the feature maps. This increases a lot the amount of trainable parameters in the following dense layer which can cause severe overfitting.
In contrast, what GAP does is to calculate the average value for each feature map across its dimensions. This comes with two advantages which are efficiency (model is lighter and faster to train) and structural regularization (preventing overfitting).

3.2.4. Classification Head and Regularization

The final part of the model is the classification head, which includes regularization and activation techniques.

A dropout layer with a rate of 0.4 was implemented after the Global Average Pooling layer with the `.Dropout(0.4)` function. During each training iteration 40% (this is where the 0.4 rate comes from) of the neurons are randomly deactivated. This is used to prevent the model from relying excessively on one feature or combination of features.

The final layer of the model is a fully-connected (Dense) layer containing a single unit

(`.Dense(1,activation="sigmoid")`). As the model has to solve a binary classification task (classifying cells into “healthy” and “ALL”) this output layer utilizes a Sigmoid activation function which returns a value between 0 and 1. This value can be interpreted as the probability of the image belonging to the ALL class so depending on whether this value is closer from 0 or 1 the model will classify the image into one of the two categories.

3.3. Training Procedure

The training strategy followed to train the model had the goal to adapt the ResNet50 architecture to the specific task of hematological imaging classification. Identifying leukemic cells requires a high degree of precision, which led to the adoption of a dual-stage training strategy. First, the backbone of the model was frozen, allowing the model to establish a stable classification head. Once this had been done, all convolutional layers were unfrozen for fine-tuning to the hematological data.

To track the models performance the following techniques were applied: a Binary Cross-Entropy loss function was utilized since the task consists in a binary classification, not only accuracy was evaluated as sensitivity and AUC (Area Under the Curve) were also monitored and the Adam optimizer was utilized due to its capability to adapt to changing learning rate.

3.3.1. Phase 1: Feature Extraction

In the first phase of training, the backbone of the ResNet50 architecture was frozen using `base_model.trainable = False`. The principal reason for this initial training phase was to train the head that was specifically built for ALL classification so it would be able to interpret the high-level features that came from the backbone, freezing the pre-trained weights from ImageNet imported with Transfer Learning to ensure they remained unchanged and therefore working as a fixed feature extractor.

In phase 1 a learning rate of 0.001 was used with the Adam optimizer. As it can be appreciated in *Figure 3* training accuracy shows a stable increase reaching a value around 85%, in contrast validation accuracy shows an unstable growth and reaching its highest value at epoch 8, at a value around 78%. This gap between the validation and training accuracy suggests an overfitting problem, highlighting the difficulty of the classification of hematological features using only the pre-trained layers. In a first attempt in building this model it was determined that earlystopping had to be used so the model would stop training if validation loss started to grow, therefore an earlystopping callback was implemented for the final model monitoring validation loss with a patience of 5 so training would stop after 5 consecutive epochs without improvement. Also the `restore_best_weights` parameter was set to True to ensure the model would keep the best weights during this phase of training avoiding the overfitting observed in the final stages of this training phase.

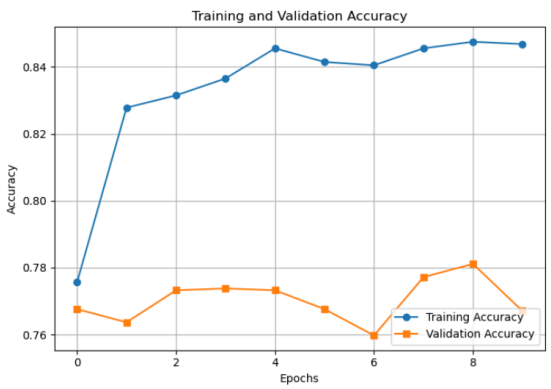

Figure 3: Plot of Training and Validation metrics for Phase 1 (accuracy and loss). Overfitting can be appreciated in the gap between the training and validation accuracy justifying the utilization of fine-tuning in Phase 2.

3.3.2. Phase 2: Fine-Tuning

After the initial head training, a fine-tuning training stage was done in an attempt to improve the model's capabilities of generalization. In phase 2 the entire backbone of the ResNet50 was unfrozen with the intention to allow all pre-trained weights to adapt to the specific textures and shapes of hematological cells. To prevent this change in training from overwriting the pre-trained weights with valuable knowledge from ImageNet a very conservative learning rate of 10^{-5} was used. This approach allowed deeper convolutional layers to adapt to the ALL classification task without losing lower-level features extraction performance.

A major part of phase 2 was addressing the class imbalance in the dataset which has already been stated in "*3.1.4. Data Distribution and Class Imbalance*". To address this class weights were set at a 2.5 to 1.0 ratio, favoring the ALL class which made the model to focus more on sensitivity by penalizing errors on leukemic cells more heavily. The decision to focus on sensitivity was made as in the medical field missing a positive case is much worse than misdiagnosing a healthy casenucas the first case can be fatal and the second one just results in a false alarm.

As shown in *Figure 4,* training accuracy hit a 100% value quickly while validation accuracy reached a value close to 85%. This rapid convergence to 100% in training accuracy is produced by the high capacity of the ResNet50 architecture to specialize in training set features, given the small size of the ALL dataset the architecture easily memorized the training data. Earlystopping was also implemented in this training phase with a patience set to 8 this time as this time training was set to reach 40 epochs.

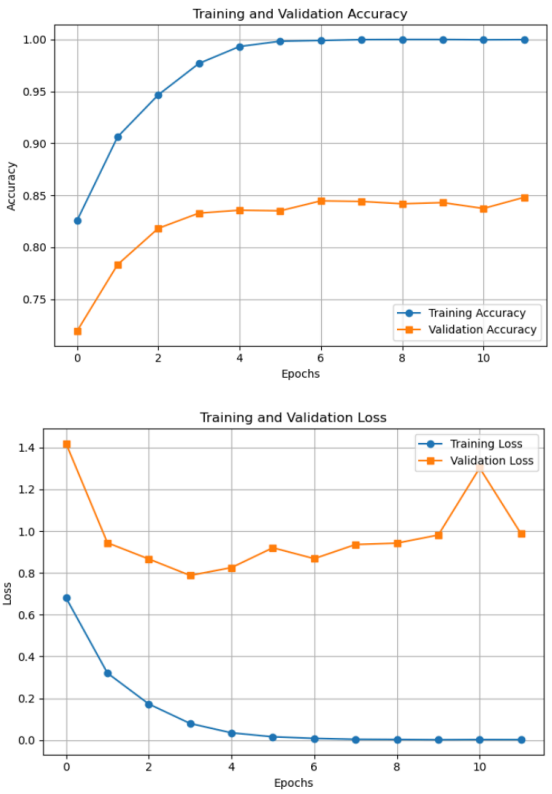

Figure 4: Plot of Training and Validation metrics for Phase 2 (accuracy and loss). While a gap between accuracies persists, the stability in validation loss and increase in accuracy compared to Phase 1 show favourable results.

4. Results

The final performance of the acute lymphoblastic leukemia classification model was evaluated using an unseen test set. This approach ensures that the model reflects true predictive metrics rather than data memorization from the training phase. After

completing the two-phase training strategy the model achieved an overall accuracy of 83.28%. This value indicates that the model is robust at this binary classification task in a hematological cell context eventhough it could be improved. However, to evaluate the diagnostic performance of the model in a clinical environment and given the class imbalance in the dataset, accuracy alone does not provide a complete picture of the diagnostic capabilities the model has. Therefore, in the following sections an analysis on class-specific metrics and a confusion matrix will be evaluated to assess the final results of the ALL classification model.

4.1. Class-Specific Analysis

To understand how the model deals with the class imbalance between healthy and leukemic samples different class-specific metrics have been analyzed including: Precision, Sensitivity and F1-Score. As shown in *Table 1* the model has a remarkably high performance at detecting healthy cells and it is also reliable when it flags a positive ALL case.

Class	Precision	Sensitivity	F1-Score	Support
Healthy	0.83	0.96	0.89	1231
ALL	0.85	0.55	0.67	546
Accuracy			0.83	1777
Macro Avg	0.84	0.76	0.78	1777
Weighted Avg	0.84	0.83	0.82	1777

Table 1: Detailed classification metrics for the validation set.

One of the results that stands out the most is the Precision of 0.8504 for the ALL class which indicates that when the model predicts a cell is leukemic it is correct the 85% of the time. This value shows that "false alarms" are minimized avoiding patients unnecessary stress of further testing in a hospital. On the other hand, a value of 0.5582 was obtained for Sensitivity for the ALL class meaning the model currently detects slightly more than half of all ALL cases. While the sensitivity value has been improved along the different models that were built to get to this one, starting at 32% in an initial model and reaching a value of 55.82% in the final one, this value highlights the difficulty in classifying the two classes given the reduced dimensions of the dataset and the morphological similarities between both healthy and leukemic cells. Finally, the F1-Score of 0.67 for the ALL class gives a balanced view of the last two metrics evaluated showing that even though the model is not yet suitable to be used in a real clinical scenario, in has successfully learned to identify leukemic features despite the healthy samples being more abundant in the dataset.

4.2. Error Analysis and Confusion Matrix

For a further evaluation of the diagnostic reliability of the model a confusion matrix was generated to visualize the distribution of correct and incorrect classifications for both categories (*Figure 5*).

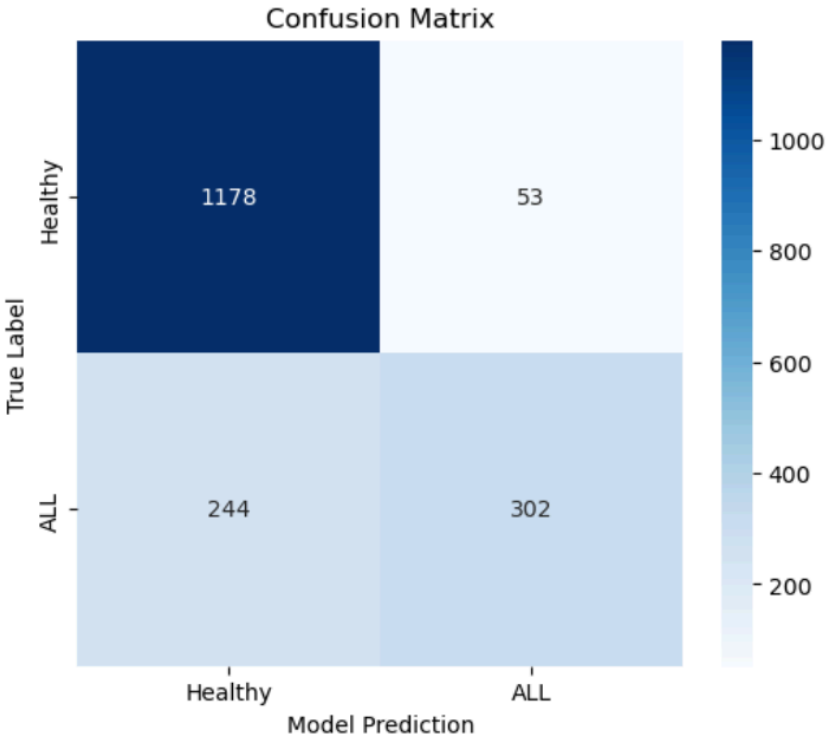

Figure 5: Confusion Matrix for the final model

As shown in *Figure 5* the model shows an exceptional ability to identify healthy cells, correctly classifying 1178 out of 1231 healthy cells which corresponds to an approximately 96% specificity highlighting that the model minimizes false positives or "false alarms" misdiagnosing only 53 healthy samples as ALL out of the total tested.

On the other hand, it can be observed that the main challenge for the model was to identify leukemic cells as out of the 546 ALL samples tested, the model correctly identified 302 samples (True Positives) and misclassified 244 as healthy (False Negatives). Even though class weights were used in phase 2 to train the model pushing it to prioritize the ALL class improving previous sensitivity values, the high number of false negatives suggests that morphological differences between healthy and leukemic cells are almost indistinguishable which could be due to low resolution or perhaps a dataset size problem.

As stated before, this model cannot be used in a real clinical scenario as false negatives still have a lot of room for improvement and a missed diagnosis has much more severe consequences than a "false alarm". The results show that the ResNet50 has the capability to specialize in hematological features but a larger and more diverse dataset would be

required to increase the overall sensitivity of the model.

5. Discussion

The results obtained after evaluating the model suggest that the implementation of the ResNet50 architecture is a viable approach to classify Acute Lymphoblastic Leukemia from blood smear images by reaching an accuracy of 83.28% and a precision of over 85% for the ALL class.

However, the gap between precision and sensitivity indicates that the models performance is heavily influenced by the training strategy and some characteristics of the dataset. The following sections discuss and clarify the decisions made along the development of the model, analyze the clinical implications of the metrics obtained, and explains the limitations of the model as well as future work plans.

5.1. Model Selection and Training Strategy

The following sections 5.1.1. Model Selection and 5.1.2. Training Strategy have the objective to justify the use of the ResNet50 architecture and the two-phase training strategy as well as explaining the thought process that lead me to that decision.

5.1.1. Model Selection

The decisions of implementing Transfer Learning and using the ResNet50 architecture were based on two main factors: subtle and difficult features had to be extracted and computational cost. To classify hematological cells where morphological differences are nearly undistinguishable a very deep network was needed to extract subtle features, the main problem when building a very deep network is vanishing gradients. In traditional deep neural networks, as the architecture becomes deeper, the gradients used during backpropagation to update the weights can tend toward zero as they are multiplied through many layers, leading to a performance saturation as weights in early layers are no longer updated making the model unable to learn effectively.

Residual Networks, and therefore ResNet50, address this through skip connections which allow the gradient to flow directly through the network. These connections skip one or more layers making sure the model learns a residual mapping instead of the full mapping allowing for gradient flow even in the deepest layers of the architecture. In the case of leukemia detection, these skip connections are the ones allowing the extraction of highly complex and abstract features without risking training performance.

To solve the computational cost problem, Transfer Learning was utilized importing the pre-trained weights from ImageNet. This ensured the model began with the ability to recognize low-level features such as shape, textures and edges without the need to train the architecture from scratch with another dataset to learn them, significantly reducing the training time required .

5.1.2. Training Strategy

Arguably, the most decisive factor in the model's evolution was the transition from Phase 1 (Feature Extraction) to Phase 2 (Fine-Tuning). For the model to adapt to the hematological images classification task, the backbone of the ResNet50 was frozen and only the custom head was trained to prevent the architecture from "forgetting" all the learning imported from ImageNet, which included low-level features recognition as it has been stated before. Although modest results were obtained in this initial phase of training it was a crucial part of the training

process, this step was necessary to prevent the so called catastrophic interference. Since the initial weights for the custom head of the model were random, their large initial errors in the early epochs would have propagated back to the entire network, potentially destroying all the learning imported through transfer learning.

Once Phase 1 was complete, fine-tuning was utilized for the final part of training. In this phase all layers from the ResNet50 architecture were unfrozen so the model could transition from recognizing general shapes to identifying hematological morphological features. To implement fine-tuning by unfreezing the whole network, a significantly low learning rate was used so deeper layers could adjust to recognizing specific nuclear textures and nucleocytoplasmic ratios of lymphoblasts without losing the ability to recognize other more basic features.
During the initial stages of model development of the ALL classifier, in the second phase of training only the last twenty convolutional layers were unfrozen for fine-tuning which improved the results obtained from the feature extraction phase but were still not good enough as a lot of precision was needed for this classification task meaning even the deepest layers had to learn and adapt to the nuclear and cytoplasmic textures.

It is important to note that class weights were exclusively implemented in the fine-tuning phase of training as introducing class weights in the initial stage of training could have led to a bias problem. By doing this, the model was able to obtain a high specificity for healthy cells while also focusing on ALL cells. Implementing class weights only in the second phase of training probably slightly limited the final sensitivity of the model but in exchange, it ensured a more balanced learning process as it is reflected in the high precision value obtained.

5.2. Clinical Implications

As shown in "*4.1. Class-Specific Analysis*" a significant gap between Precision (85%) and Sensitivity (56%) for the ALL class were obtained. In a real-world clinical setting, a diagnostic tool with these characteristics would carry critical implications, as discussed before the model does a good job at classifying healthy samples and its high precision shows that when a decision is made it is most likely to be right, the problem comes from the sensitivity value obtained for the ALL class where the model is making most errors. In the medical field, it is always preferred to misdiagnose a patient with an illness they do not actually have than misdiagnosing a sick patient as healthy which could have fatal consequences as time is gold in medicine. Therefore, due to its limited sensitivity results, the model should only be used as a complementary screening aid to assist hematologists or oncologists, rather than as a primary diagnostic tool.

5.3. Limitations and Personal Reflection

The development of this project has had several limitations which had a direct impact on the models performance. Firstly, the dataset imbalance and its reduced size, with a significantly larger proportion of healthy cells than leukemic ones which even though class weights were implemented, caused the model to struggle to generalize features of the ALL class. Secondly, the morphological similarity between healthy and leukemic cells and at different stages of lymphoblasts maduration process causes their differentiation to be difficult even for experienced oncologists. For a convolutional network such as it is the ResNet50, a much larger dataset would have

been needed for the model to master the data but medical datasets are often relatively small. The need for a larger dataset also leads to the built on common commercial laptop, which limited the ability to work with more computationally intensive architectures such as the Inception Network, with more data and longer training cycles.

However, as stated before, this project had a personal and educational perspective and the primary objective of this project was not to create a market-ready diagnostic tool but to start experimenting in building, training and optimizing a Deep Learning model from scratch. I believe the core objectives of this project have been fulfilled applying the knowledge acquired over the last months taking Andrew Ng's Deep Learning Specialization [10]. I am aware of what it needs to be done to improve the model as shown in "5. Discussion" and in many other sections of this paper, but I have decided to conclude this project here as the main objective which was educational and personal by doing this on my own have been accomplished. I am looking forward to keep learning about the ML and DL field, finishing the DL Specialization and developing and creating more complex and diverse projects in the future.

6. Conclusion

This project has served to demonstrate that Deep Learning is a viable option for the automated classification of Acute Lymphoblastic Leukemia even though the final model obtained in this project is not ready to be used. By using the ResNet50 architecture and the two-phase transfer learning training strategy, the model obtained a 85% for precision and an 83.28% accuracy value, effectively overcoming the vanishing gradients problem through the skip connections in the ResNet50 architecture. This architecture choice combined with the fine-tuning process by unfreezing the entire backbone allowed the network to adapt to nuclear textures and nucleocytoplasmic ratios. Moreover, class weights during the second phase of training resulted crucial to deal with the class imbalance in the dataset which contained a larger proportion of healthy cells than leukemic cells, making it a challenge for the network to extract features to differentiate healthy cells from leukemic ones.

The 56% sensitivity value is a clear indicator of the challenges found in distinguishing the highly similar morphologies of healthy and leukemic hematological cells in a clinical context. As stated before, this model is not ready to be used in a real medical context but if it were to be improved, it would still serve as a complementary screening tool to support hematologists but in no case to be the primary diagnostic tool.

This project has served as an extraordinary learning and enriching experience which has taught me more than any course would, it has provided my understanding of the model developing process, from data downloading and preprocessing, to hyperparameter optimization and visualizing and interpreting results. This has been my first attempt to build a model to link my Biomedical Engineering knowledge with the Deep Learning content I have been learning about in the past months to take my very first steps in the Medical AI field but definitely not the lasts as I am looking forward to keep learning and hopefully being able to complete more complex and impactful challenges in the future.

Code Availability

The complete source code, including the Jupyter Notebook with the training pipeline and data visualization, is available at: [🗎 Acute Lymphoblastic Leukemia Classifier.ipynb]

References

[1] (S/f). Bloodcancerunited.org. Recuperado el 26 de marzo de 2026, de https://bloodcancerunited.org/blood-cancer/leukemia/acute-lymphoblastic-leukemia-all

[2] *Key statistics for acute Lymphocytic leukemia (ALL)*. (2026, enero 13). Cancer.org. https://www.cancer.org/cancer/types/acute-lymphocytic-leukemia/about/key-statistics.html.

[3] Shafique, S., & Tehsin, S. (2018). Computer-aided diagnosis of acute lymphoblastic leukaemia. *Computational and Mathematical Methods in Medicine*, *2018*, 6125289. https://doi.org/10.1155/2018/6125289

[4] Huang, F., Guang, P., Li, F., Liu, X., Zhang, W., & Huang, W. (2020). AML, ALL, and CML classification and diagnosis based on bone marrow cell morphology combined with convolutional neural network: A STARD compliant diagnosis research: A STARD compliant diagnosis research. *Medicine*, *99*(45), e23154. https://doi.org/10.1097/MD.0000000000023154

[5] Fahim, Y. A., Hasani, I. W., Kabba, S., & Ragab, W. M. (2025). Artificial intelligence in healthcare and medicine: clinical applications, therapeutic.
advances, and future perspectives. *European Journal of Medical Research*, *30*(1), 848. https://doi.org/10.1186/s40001-025-03196-w

[6] He, K., Zhang, X., Ren, S., & Sun, J. (2015). Deep residual learning for image recognition. En *arXiv [cs.CV]*. http://arxiv.org/abs/1512.03385

[7] *Mahajan, P. (2020, septiembre 21). Understanding Residual Network (ResNet)architecture. Medium. https://medium.com/analytics-vidhya/understanding-resnet-architecture-869915cc2a98*

[8] Social Events. (s/f). *U-Net: Convolutional Networks for Biomedical Image Segmentation*. Uni-freiburg.de. Recuperado el 26 de marzo de 2026, de https://lmb.informatik.uni-freiburg.de/people/ronneber/u-net/

[9] *Checking your browser - reCAPTCHA*. (s/f). Kaggle.com. Recuperado el 28 de marzo de 2026, de https://www.kaggle.com/datasets/andrewmvd/leukemia-classification

[10] *Deep Learning Specialization - DeepLearning.AI*. (2020, diciembre 22). Deeplearning.Ai; DeepLearning.AI. https://www.deeplearning.ai/courses/deep-learning-specialization/